Course: DATA239 – Geometry for Computer Vision
Instructor: Prof. [redacted]
Assignment: **Final Pro ect: Write u**
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Date: December 20, 2025

Abstract

After giving a brief overview of the historical context in which Napoleon's theorem arose , this report studies the theorem through multiple relevant geometric frameworks. The theorem is introduced using classical Euclidean constructions and is then proved using visual arguments, symbolic vector algebra (using a computational algebra system, and complex numbers). We also study generalisations and applications of Napoleon's Theorem, for instance, generalisations to quadrilaterals are examined through Thébault's Theorem. Applications are discussed in the contexts of geometric calibration, robust localisation via the Fermat–Torricelli point, and iterative shape regularisation through the Petr–Douglas–Neumann theorem.

1 Introduction

1.1 Historical Context

Napoleon Bonaparte's engagement with mathematics extended beyond amateur interest, serving as a professional necessity during his career as an artillery officer where applied geometry and ballistics were critical to tactical success [3]. His proficiency was formally recognized through his election to the mathematics section of the *Institut de France* in 1797, a rare distinction for a head of state [2]. Furthermore, his administration was characterized by the active patronage of scientific rationality; he cultivated close intellectual relationships with leading figures such as Gaspard Monge and Pierre-Simon Laplace, elevating them to high government positions and fostering an environment of significant mathematical advancement in France.

Despite the theorem bearing his name, historical consensus suggests that the attribution is apocryphal and that Napoleon was likely not the original discoverer [3]. Historians postulate that the theorem was actually formulated by Lorenzo Mascheroni, a contemporary Italian mathematician and friend of Bonaparte. It is believed that Mascheroni demonstrated this elegant result to Napoleon, and the theorem subsequently acquired the Emperor's name due to his immense fame and well-known association with the mathematical community of his era [3].

1.2 Stating Napoleon's Theorem

Initial Geometric Setup: We begin with an arbitrary triangle $\triangle ABC$ in the two-dimensional Euclidean plane. No restrictions are imposed on the shape of this triangle: it may be scalene, isosceles, acute, right-angled, or obtuse. This generality is essential, and

Napoleon's Theorem asserts a conclusion that is independent of the specific geometry of the original triangle. Every step that follows relies only on the side lengths of this triangle and on standard Euclidean constructions.

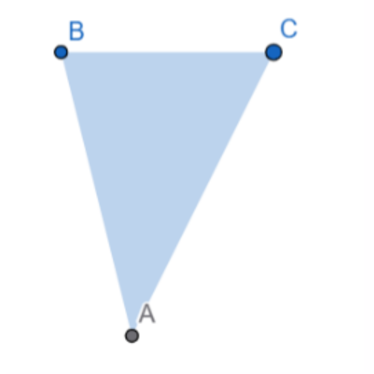

Figure 1: An arbitrary triangle ABC serving as the base configuration.

Construction on Side BC: Let $\triangle ABC$ be an arbitrary triangle in the Euclidean plane. On the side BC, we construct an equilateral triangle externally to $\triangle ABC$. Denote this equilateral triangle by $\triangle BCD$, where the vertex D lies outside the segment BC. We now determine the centroid of $\triangle BCD$. Recall that the centroid of a triangle is the unique point at which its three medians intersect, and that it divides each median in the ratio $2:1$, with the longer segment adjacent to the vertex. Let this centroid be denoted by G_2. This point will later serve as one of the vertices of the triangle associated with Napoleon's Theorem.

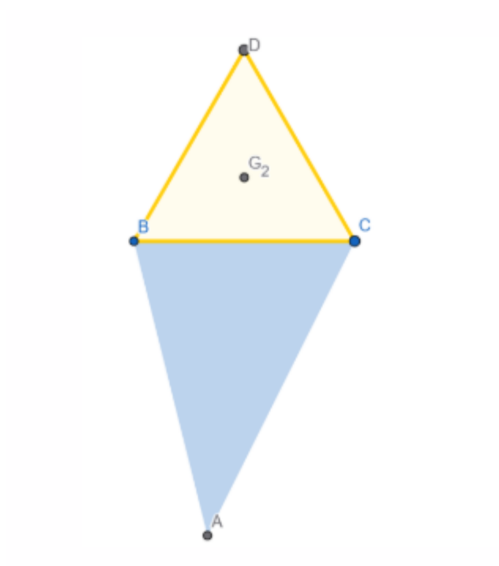

Figure 2: Equilateral triangle constructed externally on side BC and its centroid G_2.

Construction on Side AB: We repeat the same construction on side AB. An equilateral triangle is erected externally on AB, and we denote this triangle by $\triangle ABF$, with F lying outside the original triangle. The centroid of $\triangle ABF$ is then located and labeled as G_1. As in the previous case, G_1 represents the geometric center of the equilateral triangle constructed on side AB. At this stage, two centroids have been identified: G_1 corresponding to side AB, and G_2 corresponding to side BC.

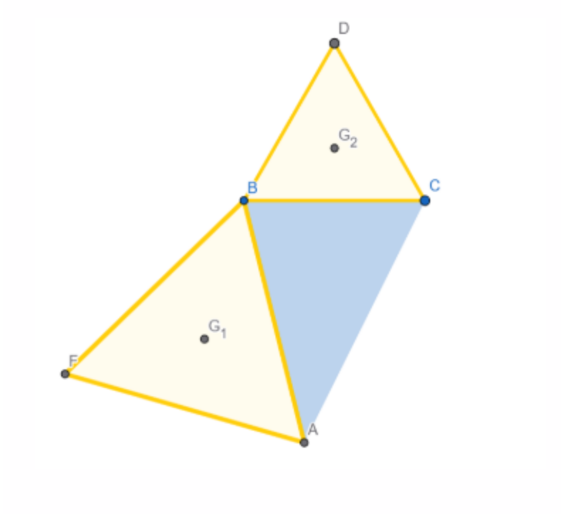

Figure 3: Equilateral triangle constructed externally on side AB and its centroid G_1.

Construction on Side AC: Finally, an equilateral triangle is constructed externally on side AC. This triangle is denoted by $\triangle ACE$, with E lying outside $\triangle ABC$. The centroid of $\triangle ACE$ is identified and marked as G_3. With this construction, each side of the original triangle now has an associated equilateral triangle and a corresponding centroid. This completes the symmetric geometric construction on all three sides of $\triangle ABC$.

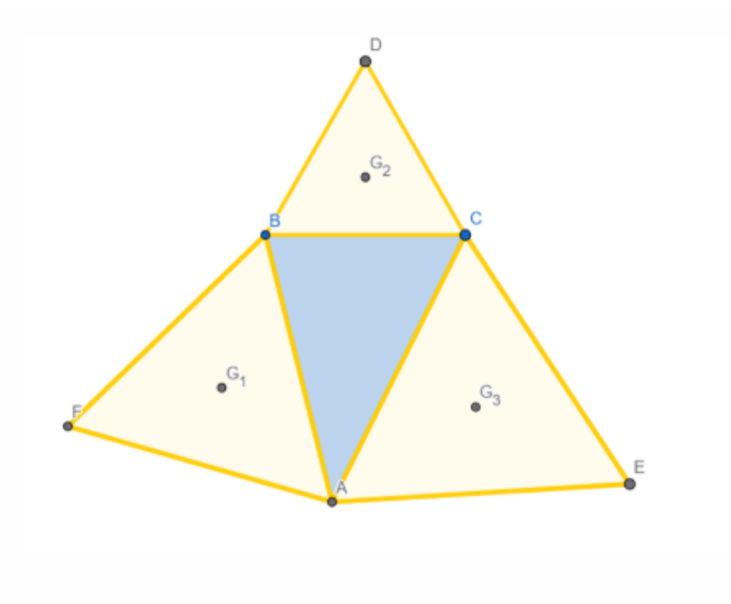

Figure 4: Equilateral triangle constructed externally on side AC and its centroid G_3.

The Three Centroids: We now focus on the three points G_1, G_2, and G_3, each of which is the centroid of an equilateral triangle constructed externally on a side of $\triangle ABC$. Specifically:

- G_1 is the centroid of the equilateral triangle on side AB,
- G_2 is the centroid of the equilateral triangle on side BC,
- G_3 is the centroid of the equilateral triangle on side AC.

These points arise purely from geometric constructions based on the original triangle, yet none of them lies on $\triangle ABC$ itself. The central claim of Napoleon's Theorem concerns the geometric relationship among these three centroids.

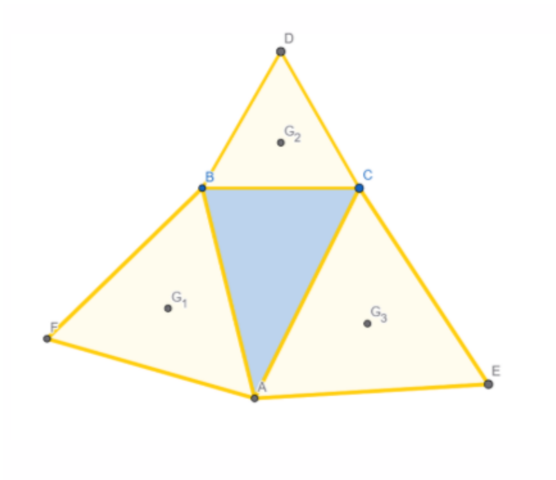

Figure 5: The three centroids G_1, G_2, and G_3 associated with the equilateral constructions.

Statement of Theorem: Napoleon's theorem states that if equilateral triangles are constructed externally on the sides of any triangle, then the centroids of those equilateral triangles themselves form an equilateral triangle. In the notation used above, this means that the triangle $\triangle G_1G_2G_3$ is equilateral.

What makes this result particularly striking is that it holds independently of the size, orientation, or shape of the original triangle $\triangle ABC$. We have crafted visualisations on GeoGebra Geometry software (linked in Supplementary document) that illustrate this.

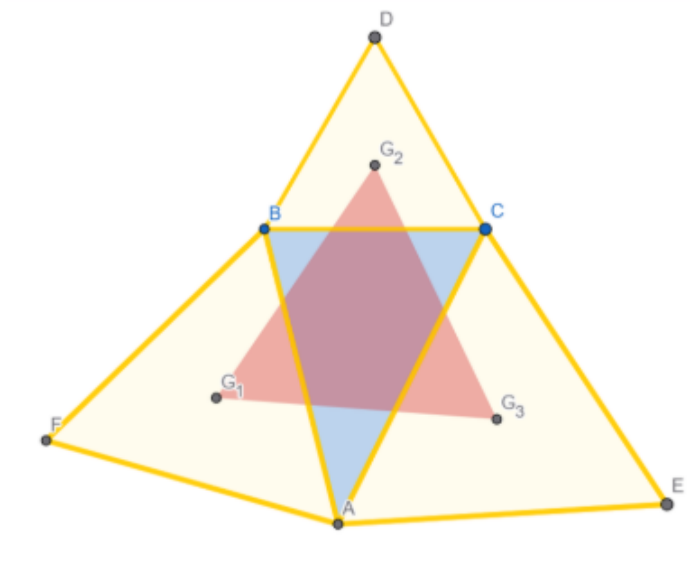

Figure 6: The Napoleon triangle formed by joining G_1, G_2, and G_3.

2 Proving Napoleon's Theorem

This section focuses on presenting proofs to Napoleon's theorem using a wide range of techniques, including constructions, symbolic methods as well as working with complex numbers. The figures in this section were produced using the GeoGebra software.

2.1 A Visual Proof

Initial Configuration: We begin by defining an arbitrary triangle, denoted $\triangle ABC$. In order to preserve the full generality of the theorem, the vertices A, B, and C are chosen so that the resulting triangle is scalene and exhibits no inherent symmetries such as equal sides or angles. As illustrated in the corresponding figure, the side lengths and internal angles are all distinct, providing a suitable configuration for testing the invariance of the theorem.

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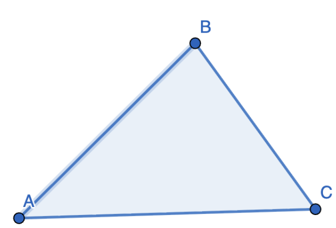


Figure 7: An arbitrary scalene triangle ABC used as the base configuration.

Exterior Equilateral Constructions: Following the construction prescribed by Napoleon's Theorem, equilateral triangles are erected externally on each side of the base triangle $\triangle ABC$. Specifically, an equilateral triangles $\triangle ADB$, $\triangle AEC$, $\triangle BFC$ are constructed on sides AB, AD and AC respectively. From each of these exterior equilateral triangles, the centroid is determined. These centroids are G, H and I for an equilateral triangles $\triangle ADB$, $\triangle AEC$ and $\triangle BFC$ respectively.

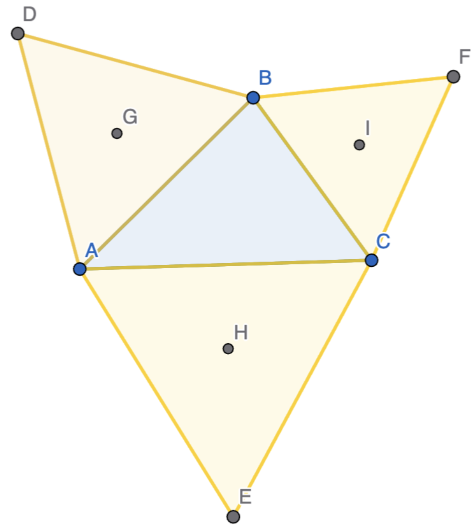


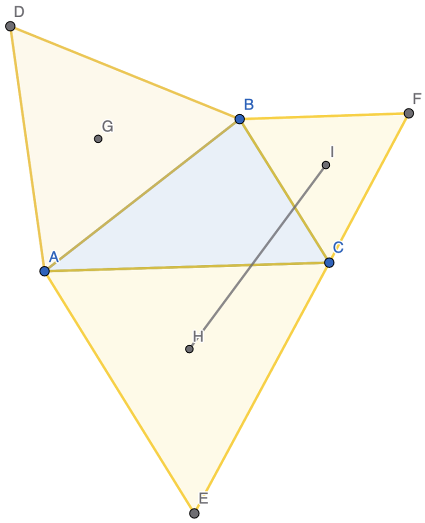
Figure 8: Construction of exterior equilateral triangles on the sides of the arbitrary scalene triangle ABC, showing the identified centroids G, H, and I.

Manipulating figure using Isometries: The triangles in the figure are related by a rigid motion of the plane. Specifically, the triangle $AB'C'$ is obtained from $\triangle ABC$ by a rotation of $120°$ about the point H indicated in the diagram.

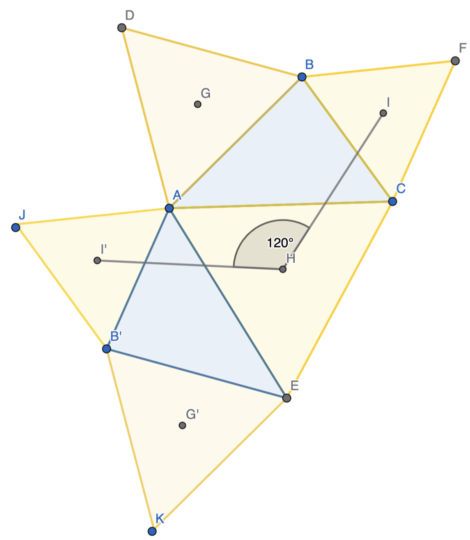
Under this rotation, the line segment IH maps to $I'H$, and each vertex of $\triangle ABC$ is carried to the corresponding primed vertex:$B \mapsto B', \quad C \mapsto E$. The orientation of the triangle is preserved, and all side lengths and angles remain unchanged.

The same rotation carries each of the auxiliary triangles sharing sides with $\triangle ABC$ to their corresponding congruent copies. Consequently, all shaded triangles in the configuration are pairwise congruent and arranged cyclically around the common center H.

Thus, the entire figure is generated by a $120°$ rotational symmetry about H.



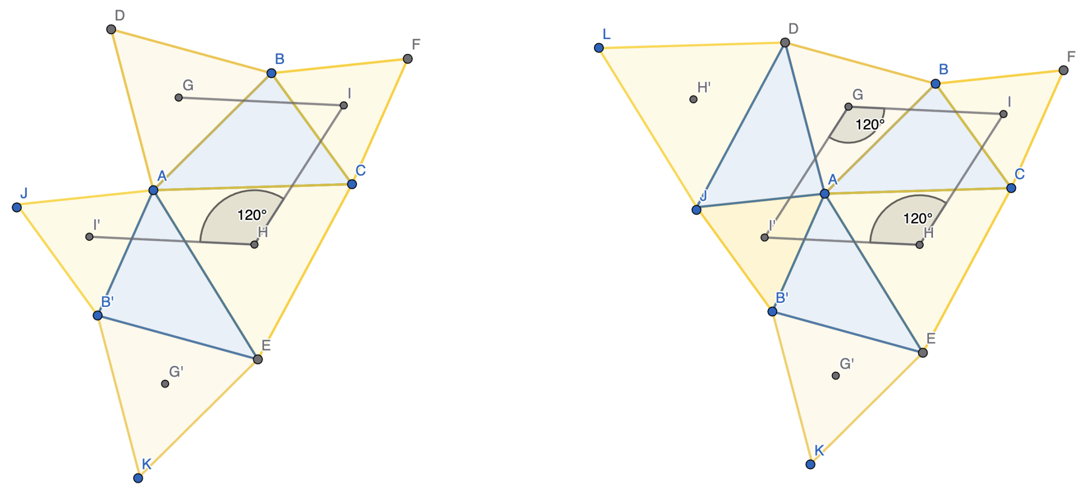
Before Isometry: Joining IH to illustrate rotation



After Isometry

Figure 9: Rotating the original triangle and the original constructed equilateral triangles by $120°$ about H. The segment IH maps to HI'

Now, $IH = I'H$, as this was an isometric transformation.
We can now do a similar isometric transformation, where we can rotate the original triangle and the constructed equilateral triangles GI about G by $120°$, such that $GI \mapsto GI'$.



Before Transformation

After Transformation

Figure 10: Rotating the original triangle and the original constructed equilateral triangles by 120° about G. The segment GI maps to GI', forming a parallelogram

Similarly, $GI = GI'$, since the transformation is an isometry. This implies that the quadrilateral $IGI'H$ is a parallelogram with equal sides. Thus, when we construct diagonal GH, it bisects $\angle IGI'$ and $\angle IHI'$, as in the following image.

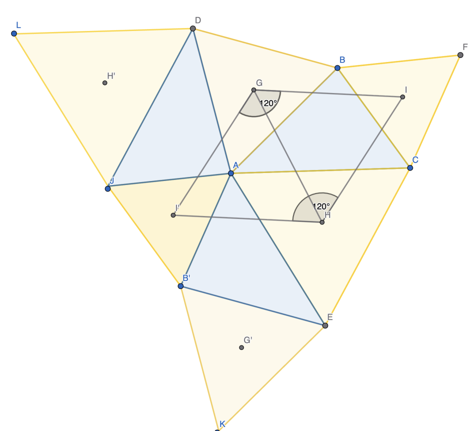


Figure 11: Constructing Diagonal GH

Thus, $\angle IGH = \angle IHG = 60°$. Hence, due to the property of sum of interior angles of a triangle, $\angle GIH = 60°$, which proves that $\triangle IGH$ is equilateral, which concludes our visual proof.

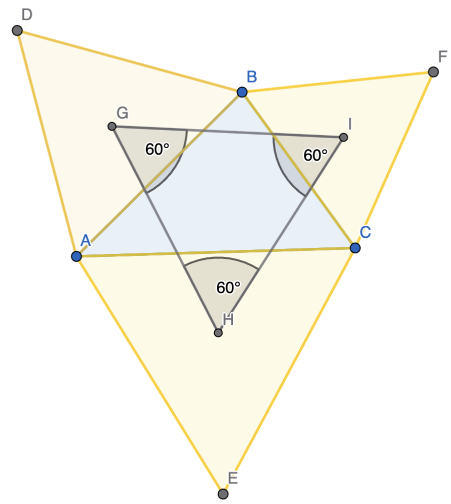


Figure 12: The triangle formed by joining centroids G, I, and H, verified to be equilateral – Figure generated by hiding Isometric constructions

2.2 Vector-based Symbolic Proof of Napoleon's Theorem

We now present a vector formulation of Napoleon's Theorem, closely mirroring the geometric construction while allowing symbolic verification via linear algebra. All computations are carried out in $\mathbb{R}^2$, with a Computational Algebra System (Mathematica) used to confirm algebraic identities.

Step 1: Normalising the configuration

Let the vertices of $\triangle ABC$ be vectors

$$A = (x_1, y_1), \qquad B = (x_2, y_2).$$

Since translation is an isometry, we may assume without loss of generality that the centroid of $\triangle ABC$ lies at the origin. The centroid condition

$$\frac{A + B + C}{3} = 0$$

implies

$$C = -A - B.$$

We do this as placing the centroid at the origin removes complications due to translational degrees of freedom. This simplification is valid as all metric properties relevant to the theorem are invariant under the transformation.

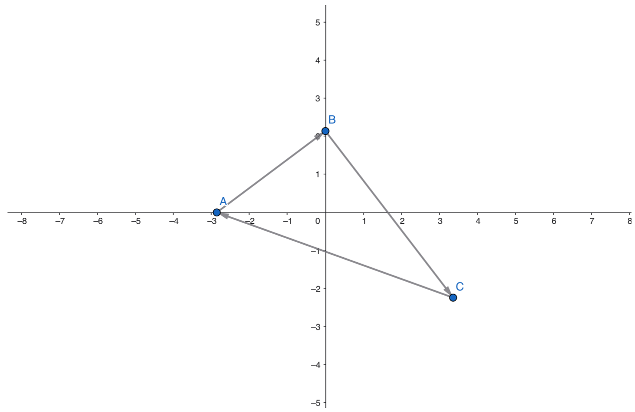


Figure 13: Arbitrary triangle ABC in $\mathbb{R}^2$ with centroid at the origin

Code (Mathematica):

```
a = {x1, y1};
b = {x2, y2};
c = -a - b;
Print[c]
```

Output:

```
{-x1 - x2, -y1 - y2}
```

Step 2: Constructing the equilateral triangles

To construct equilateral triangles externally on each side, we rotate the corresponding side vectors by $60°$ counterclockwise. In $\mathbb{R}^2$, this is achieved using the rotation matrix

$$R_{60} = \begin{pmatrix} \cos(\pi/3) & -\sin(\pi/3) \\ \sin(\pi/3) & \cos(\pi/3) \end{pmatrix}.$$

Consider the side BC. The equilateral triangle $\triangle BCA_1$ constructed externally on BC has its third vertex A_1 obtained from point vector B by adding the side vector $C-B$, rotated by $60°$: $A_1 = B + R_{60}\,(C - B)$.

Similarly, for the equilateral triangles on the other sides we have

$$B_1 = C + R_{60}\,(A - C), \qquad C_1 = A + R_{60}\,(B - A).$$

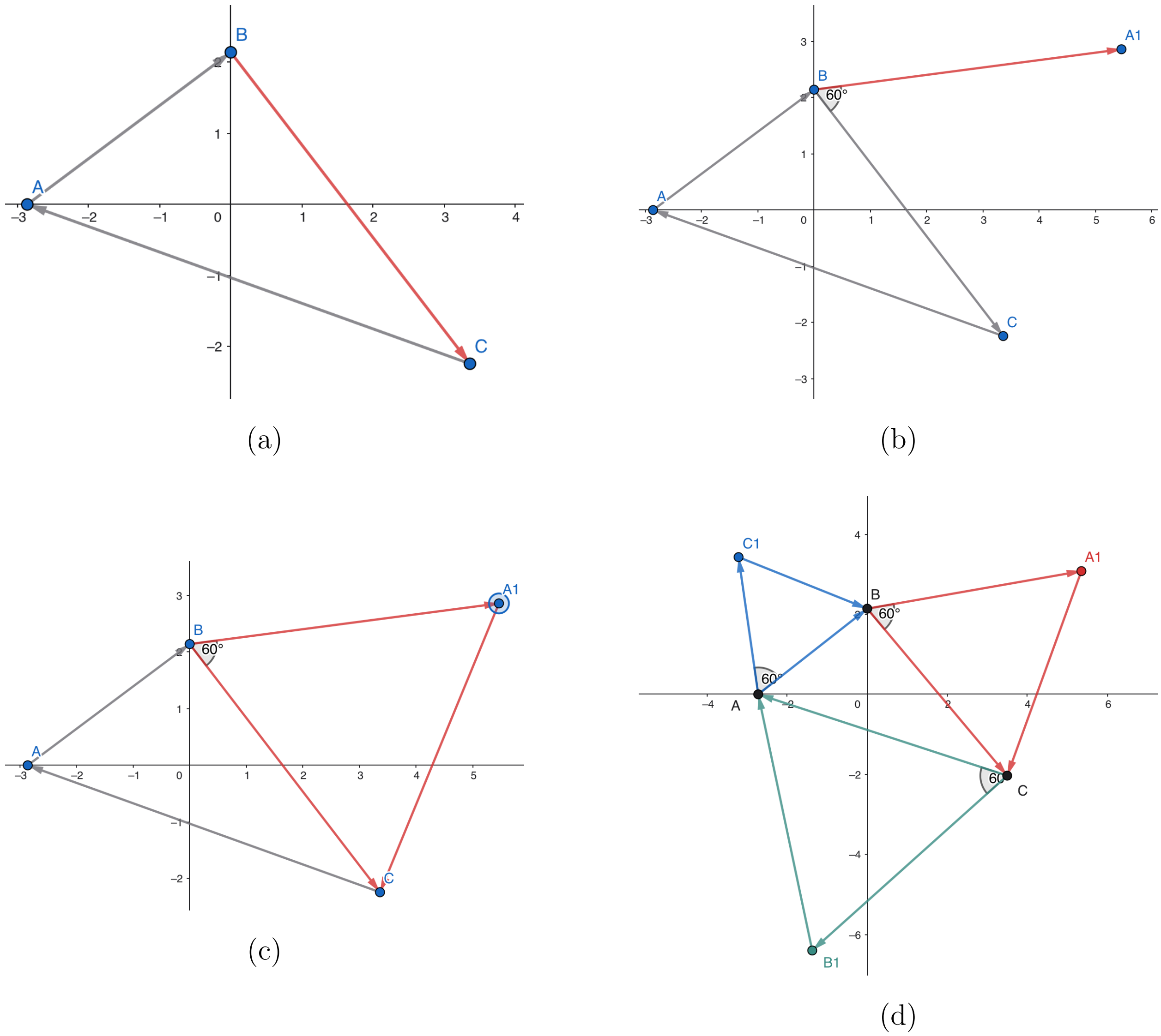


Figure 14: Constructing Equilateral Triangles using pertinent vector additions and rotations

Code (Mathematica):

```
R60 = {{Cos[Pi/3], -Sin[Pi/3]},
       {Sin[Pi/3],  Cos[Pi/3]}};

a1 = b + R60.(c - b);
b1 = c + R60.(a - c);
c1 = a + R60.(b - a);
```

Step 3: Centroids of the equilateral triangles

Let O_a, O_b, and O_c denote the centroids of triangles $\triangle A_1BC$, $\triangle AB_1C$, and $\triangle ABC_1$, respectively. Each centroid is given by the arithmetic mean of its vertices:

$$O_a = \frac{A_1 + B + C}{3}, \quad O_b = \frac{A + B_1 + C}{3}, \quad O_c = \frac{A + B + C_1}{3}.$$

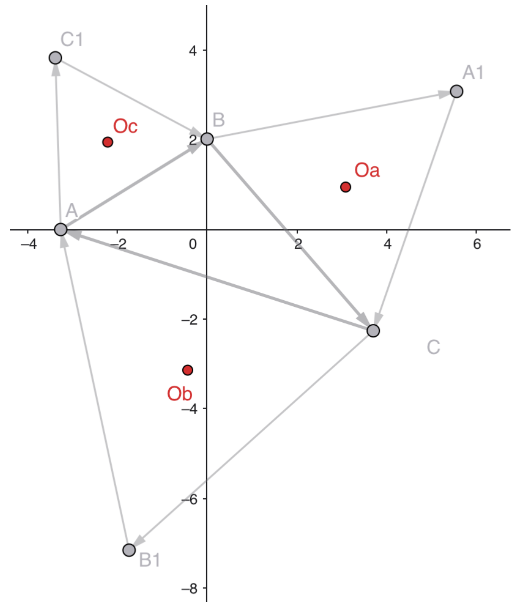


Figure 15: The centroids of the constructed triangles.

Code (Mathematica):

```
oa = (a1 + b + c)/3;
ob = (a + b1 + c)/3;
oc = (a + b + c1)/3;
```

Step 4: Side vectors of the Napoleon triangle

To test whether $\triangle O_aO_bO_c$ is equilateral, we compare the lengths of its sides. These are represented by the difference vectors:

$$\overrightarrow{O_aO_b} = O_b - O_a, \quad \overrightarrow{O_aO_c} = O_c - O_a, \quad \overrightarrow{O_bO_c} = O_b - O_c.$$

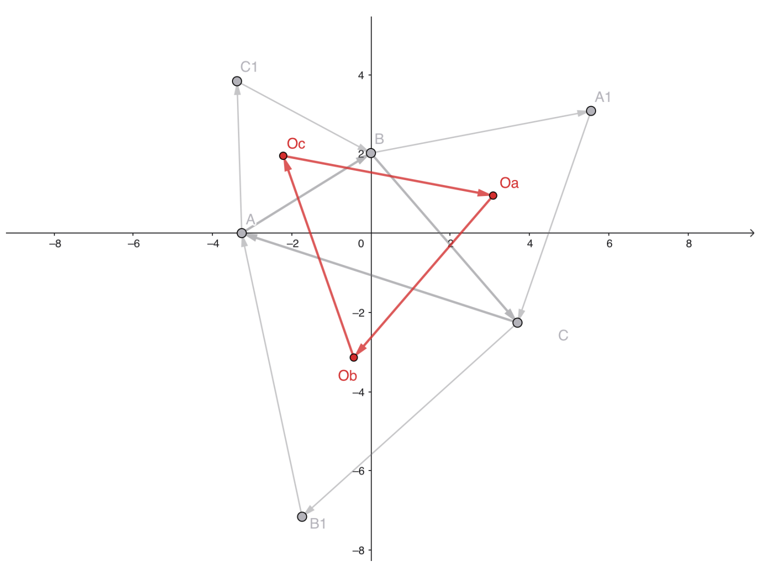


Figure 16: Constructing the Napoleon triangle.

Code (Mathematica):

```
oaob1 = ob - oa;
oaoc1 = oc - oa;
oboc1 = ob - oc;
```

Step 5: Symbolic verification of equal lengths

We compute the Euclidean norms of the side vectors and verify their equality symbolically.

Code (Mathematica):

```
exp1 = Norm[oaob1];
exp2 = Norm[oaoc1];
exp3 = Norm[oboc1];

FullSimplify[exp1 == exp2]
FullSimplify[exp1 == exp3]
FullSimplify[exp2 == exp3]
```

Output:

```
True
True
True
```

Conclusion: The symbolic equalities confirm that $|O_aO_b| = |O_bO_c| = |O_cO_a|$, and hence $\triangle O_aO_bO_c$ is equilateral, which completes the symbolic vector proof for Napoleon's Theorem.

2.3 Napoleon's Theorem: A Proof with Complex Numbers

For this proof, we adopt a similar approach and nomenclature to the vector proof described above, but continue to present it without assuming a background in the previous proof for coherence. This is a proof similar to many presented in the course, in the sense that we apply transformations that preserve invariant properties to work with the problem in a more mathematically convenient modality, and extrapolate the findings to a more general statement. Parts of this proof were adapted from [1].

2.3.1 Notation and Conventions

Working with complex numbers is equivalent to working with vectors in $\mathbb{R}^2$ under the transformation T_c below.

We define the identification map $T_c : \mathbb{R}^2 \to \mathbb{C}$ by $T_c(x, y) = x + iy$. Under this identification, adding complex numbers is equivalent to adding vectors in $\mathbb{R}^2$.

Rotation: $\mathbb{R}^2$ versus $\mathbb{C}$. Rotating vectors in $\mathbb{R}^2$ by an angle θ is given by the standard rotation matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

In the complex plane, we represent the same rotation by multiplication with

$$M_e(\theta) = e^{i\theta} = \cos\theta + i\sin\theta.$$

Algebraic check: $T_c \circ R_\theta = M_e(\theta) \circ T_c$. Let $v = (x, y) \in \mathbb{R}^2$. Then

$$(R_\theta v) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta),$$

so applying T_c gives

$$\begin{aligned}
(T_c \circ R_\theta)(x, y) &= T_c(x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta) \\
&= (x\cos\theta - y\sin\theta) + i(x\sin\theta + y\cos\theta) \\
&= (x\cos\theta + ix\sin\theta) + (-y\sin\theta + iy\cos\theta) \\
&= x(\cos\theta + i\sin\theta) + y(-\sin\theta + i\cos\theta) \\
&= x(\cos\theta + i\sin\theta) + y\, i(\cos\theta + i\sin\theta) \\
&= (\cos\theta + i\sin\theta)(x + iy) \\
&= e^{i\theta}\, T_c(x, y) \\
&= (M_e(\theta) \circ T_c)(x, y).
\end{aligned}$$

Hence, for all $v \in \mathbb{R}^2$,

$$T_c \circ R_\theta(v) \ = \ M_e(\theta) \circ T_c(v).$$

Thus, to rotate a vector z in the complex plane by an angle θ, we multiply z by $e^{i\theta}$.

Why use complex numbers:

Identifying the Euclidean plane with the complex plane $\mathbb{C}$ allows planar geometry to be handled using a single algebraic object per point. Geometric transformations that would otherwise require matrix or vector calculus can then be expressed as elementary algebraic operations on complex numbers. For instance, as shown above, translations and rotations are represented by addition and multiplication in $\mathbb{C}$. As a result, compositions of rigid motions reduce to simple products and sums, avoiding repeated coordinate or matrix manipulations. This makes it particularly effective for theorems where symmetry and rotation play a central role, such as Napoleon's and Van Aubel's Theorems as well as other generalisations. In the supplementary section of this assignment, we also present a proof on Van Aubel's Theorem.

Proof. We 'identify' the Euclidean plane with the complex plane $\mathbb{C}$: a point (x, y) is represented by $z = x + iy$. Translations correspond to addition of complex numbers, and a rotation by angle θ about the origin corresponds to multiplication by $e^{i\theta}$.

Step 1: Normalising the triangle. Let $P, Q, R \in \mathbb{C}$ be three 'non-collinear' complex numbers corresponding to the vertices of $\triangle PQR$. Translate the whole configuration $\triangle PQR$ to correspond to $\triangle ABC$ so centroid of $\triangle ABC$ is at the origin. This is in line with the condition in the vector proof above and does not change any shape or angle, so it preserves the statement of the theorem.

The centroid condition is

$$\frac{A + B + C}{3} = 0 \quad \Longrightarrow \quad A + B + C = 0.$$

Step 2: Constructing the equilateral triangles. Let $d = e^{i\pi/3}$, so that multiplication by d is a counterclockwise rotation by $60°$. Because $|d| = 1$, this is a pure rotation (no scaling).

As above, the third vertices of the equilateral triangles are then obtained by rotating each side vector about the appropriate base point:

$$A_1 = B + d(C - B), \quad B_1 = C + d(A - C), \quad C_1 = A + d(B - A).$$

This construction encodes the geometric rule "rotate the side by $60°$ about one endpoint" directly in our complex number schema.

Step 3: Coordinates of the centroids. By definition, each centroid is the average of the three vertices of the corresponding equilateral triangle.

For $\triangle BCA_1$:

$$O_a = \frac{B + C + A_1}{3} = \frac{B + C + B + d(C - B)}{3} = \frac{2B + C + d(C - B)}{3}.$$

For $\triangle CAB_1$:

$$O_b = \frac{C + A + B_1}{3} = \frac{C + A + C + d(A - C)}{3} = \frac{2C + A + d(A - C)}{3}.$$

For $\triangle ABC_1$:

$$O_c = \frac{A + B + C_1}{3} = \frac{A + B + A + d(B - A)}{3} = \frac{2A + B + d(B - A)}{3}.$$

Step 4: Side vectors of $\triangle O_aO_bO_c$. We now compute the vectors corresponding to the sides of $\triangle O_aO_bO_c$.

First,

$$\begin{aligned}\overrightarrow{O_aO_b} &= O_b - O_a \\ &= \frac{2C + A + d(A - C) - (2B + C + d(C - B))}{3} \\ &= \frac{A - 2B + C + d(A - C) + d(B - C)}{3}.\end{aligned}$$

Using $A + B + C = 0$, equivalently $C = -A - B$, one simplifies to

$$\overrightarrow{O_aO_b} = -B - dC.$$

Analogously,

$$\overrightarrow{O_bO_c} = -C - dA, \qquad \overrightarrow{O_cO_a} = -A - dB.$$

Step 5: Showing that the sides differ by a $120°$ rotation. We compare $\overrightarrow{O_bO_c}$ and $\overrightarrow{O_cO_a}$. From above,

$$\overrightarrow{O_cO_a} = -A - dB.$$

Multiplying by d^2 (a $120°$ rotation since $d^2 = e^{i2\pi/3}$), we get

$$d^2\,\overrightarrow{O_cO_a} = d^2(-A - dB) = -d^2A - d^3B.$$

But $d = e^{i\pi/3}$ satisfies $d^6 = 1$, hence $d^3 = -1$, so

$$d^2\, \overrightarrow{O_cO_a} = -d^2 A + B.$$

On the other hand,

$$\overrightarrow{O_bO_c} = -C - dA = -(-A - B) - dA = A + B - dA,$$

since $C = -A - B$.

Thus we want to check that

$$A + B - dA = -d^2 A + B,$$

or equivalently

$$A(1 - d) = -d^2 A.$$

This holds because d satisfies

$$d^2 - d + 1 = 0 \quad \Longrightarrow \quad 1 - d = -d^2.$$

Therefore

$$\overrightarrow{O_bO_c} = d^2\, \overrightarrow{O_cO_a}.$$

In words: the side $\overline{O_bO_c}$ is obtained from $\overline{O_cO_a}$ by a counterclockwise rotation of 120° about the origin.

By the same type of computation, one finds that each side of $\triangle O_aO_bO_c$ is obtained from the previous one by multiplying by $d^2 = e^{i2\pi/3}$. Hence, successive side vectors differ by a 120° rotation. As $|d^2| = 1$, the lengths of all three side vectors are equal.

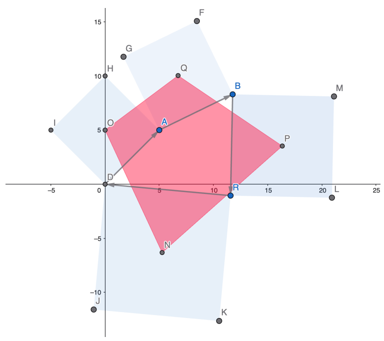
Thus $\triangle O_aO_bO_c$ is equilateral, which proves Napoleon's Theorem.

□

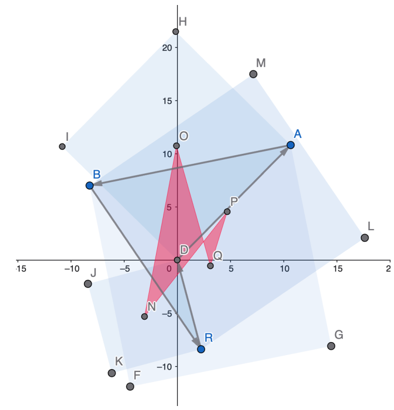
3 Generalisations and Applications

3.1 A Generalisation to Quadrilaterals

While Napoleon's Theorem guarantees that constructing equilateral triangles on the sides of any triangle results in a regular triangle, a direct equivalent does not hold for arbitrary quadrilaterals. As shown in Figure 17, when squares are erected on the sides of a random quadrilateral, the shape formed by their centres (shown in red) is generally not a square [4].



(a)

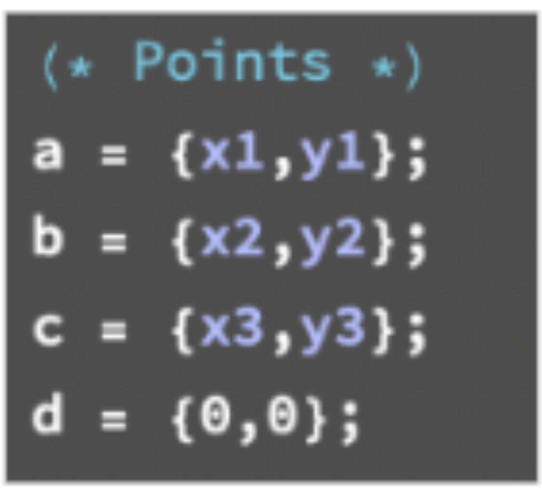


(b)

Figure 17: Failure to generalise to arbitrary quadrilaterals as shown in panels (a) and (b).

3.1.1 Deriving Constraints via CAS

To determine if specific conditions exist under which the theorem holds, we utilised Mathematica. Similar to the vector proof above, we can define the vertices of the quadrilateral symbolically and solve for the coordinates that satisfy the regularity condition. We were able to derive a set of constraints, and have attached the Mathematica code in the supplementary file. As illustrated in Figure 18, these constraints are generally quite complex, involving intricate relationships between the coordinates of the vertices (x_1, y_1, x_2, y_2, etc.).



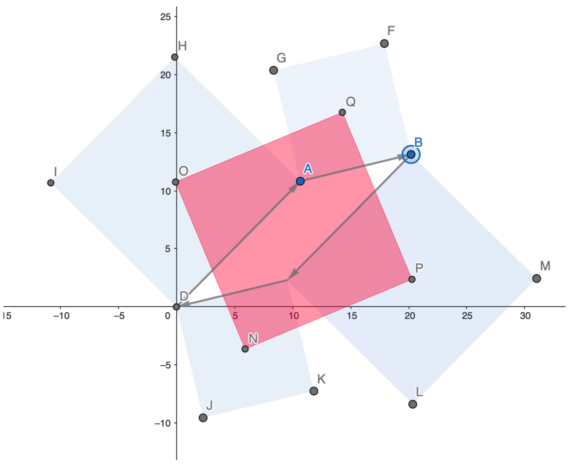
(a) Definition of points.

(b) Complex constraints derived via CAS.

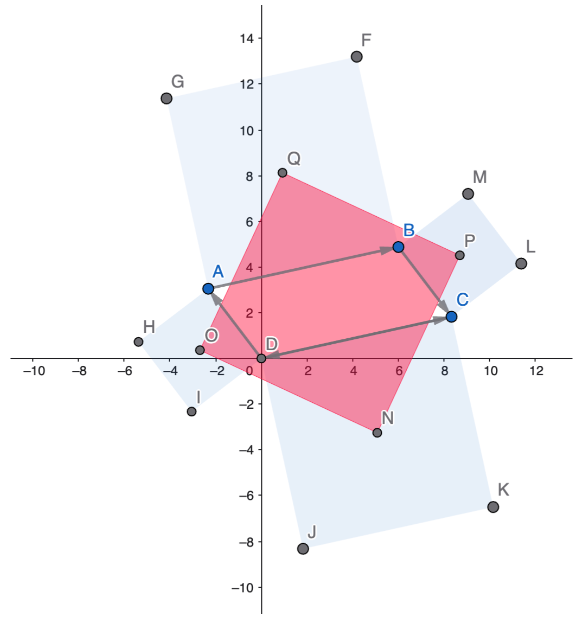
Figure 18: Using CAS to find the algebraic constraints for the theorem to hold.

3.1.2 Thébault's Theorem

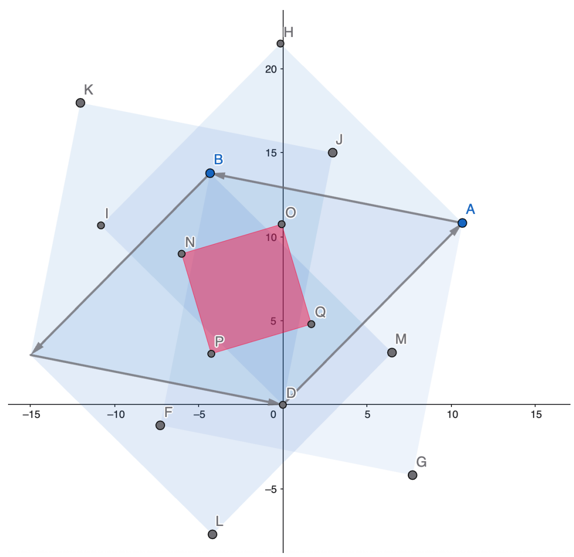
When one of these complex constraints is simplified, a significant special case emerges. We observe that if the initial quadrilateral is a **parallelogram**, the centers of the squares constructed on its sides do indeed form a square. This result is classically known as **Thébault's Theorem**.



(a)



(b)



(c)

Figure 19: Quadrilateral analogue of Napoleon's Theorem – Illustrating Thébault's Theorem.

3.1.3 Relation to Napoleon's Theorem

Thébault's Theorem serves as a quadrilateral analogue to Napoleon's Theorem.

- **Napoleon:** Takes a triangle (3 sides) $\rightarrow$ constructs triangles $\rightarrow$ yields a Regular Triangle.

- **Thébault:** Takes a parallelogram (4 sides) → constructs squares → yields a Regular Quadrilateral (Square).

While Napoleon's theorem is robust for *any* triangle, Thébault's theorem illustrates that for $n = 4$, we must restrict the input shape (to a parallelogram) to guarantee a regular output. This was verified using Mathematica.

3.2 The Inner and Outer Napoleon Triangles

Beyond the shape regularity, Napoleon's Theorem offers a powerful metric regarding area. We can construct two distinct shapes:

1. **Outer Napoleon Triangle:** Constructed by erecting equilateral triangles pointing *outward*.
2. **Inner Napoleon Triangle:** Constructed by erecting equilateral triangles pointing *inward* (overlapping the original triangle).

3.2.1 The Area Difference Theorem

A fundamental property links these two constructions to the original shape. The difference between the area of the outer Napoleon triangle (A_{out}) and the inner Napoleon triangle (A_{in}) is exactly equal to the area of the original triangle (A_{orig}):

$$\text{Area}_{\text{outer}} - \text{Area}_{\text{inner}} = \text{Area}_{\text{original}}$$

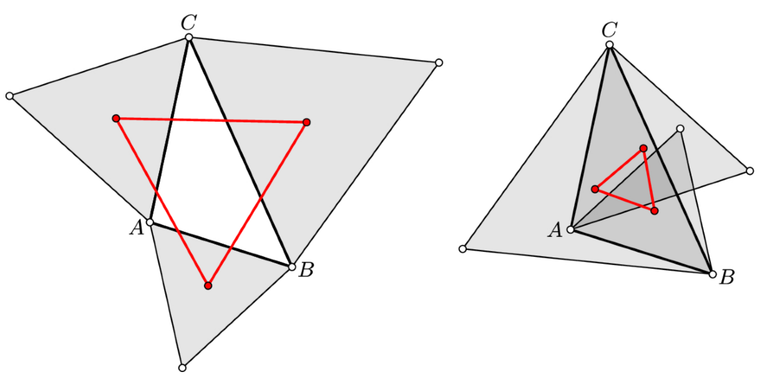


Figure 20: Visualising the Outer (blue) and Inner (red, dashed) Napoleon Triangles. The difference in their areas equals the area of the original black triangle.

3.2.2 Relation to Napoleon's Theorem

This theorem complements the original geometric statement. While the standard theorem proves *shape* invariance (regularity), this corollary proves *magnitude* invariance. It demonstrates that the "Napoleon transformation" conserves area information strictly, regardless of the distortion of the original triangle.

3.2.3 Application: Calibration and Reference

In computer vision calibration, this invariant provides a robust error-checking mechanism. If a system detects a triangle and computes the Napoleon areas, any deviation from the equation $A_{out} - A_{in} = A_{orig}$ indicates non-linear distortion (e.g., lens warping or perspective skew). The conserved area serves as a "ground truth" for calibrating geometric measurement tools.

3.3 The Fermat-Torricelli Point: Optimisation & Localisation

The Fermat-Torricelli point acts as a bridge between pure geometry and optimisation theory. It is the geometric median of a triangle, defined as the point minimising the sum of distances to the vertices.

3.3.1 Relation to Napoleon's Theorem

The Fermat point is "genetically" identical to Napoleon's Theorem. Both proofs rely on the same construction: erecting equilateral triangles on the sides of the base triangle.

- **Napoleon:** Connects the *centers* of these new triangles to find a regular shape.
- **Fermat:** Connects the *vertices* of these new triangles to the opposite corners of the original triangle (e.g., vertex C to the tip of the triangle on AB).

Thus, the Fermat point application is essentially using the "Napoleon scaffolding" for optimisation rather than shape generation.

3.3.2 Mathematical Proof: The Rotation Isometry

The standard proof relies on a 60° rotation, creating a configuration identical to the external equilateral triangles in Napoleon's Theorem.

1. **Construction:** Let P be a point inside $\triangle ABC$. Rotate the segment AP and the triangle $\triangle APC$ by 60° counter-clockwise around vertex A to obtain $\triangle AP'C'$.
2. **Isometry Properties:**
   - Since rotation is rigid, length is preserved: $PC = P'C'$.
   - By construction, $\triangle APP'$ is isosceles with a 60° angle, making it equilateral. Thus, $PA = PP'$.
3. **Path Transformation:** The total distance $D = PA + PB + PC$ can now be rewritten as a continuous path:
$$D = BP + PP' + P'C'$$
4. **Optimization:** The shortest path between two points (B and C') is a straight line. Therefore, the minimum occurs when B, P, P', C' are collinear, as illustrated in Figure 21.

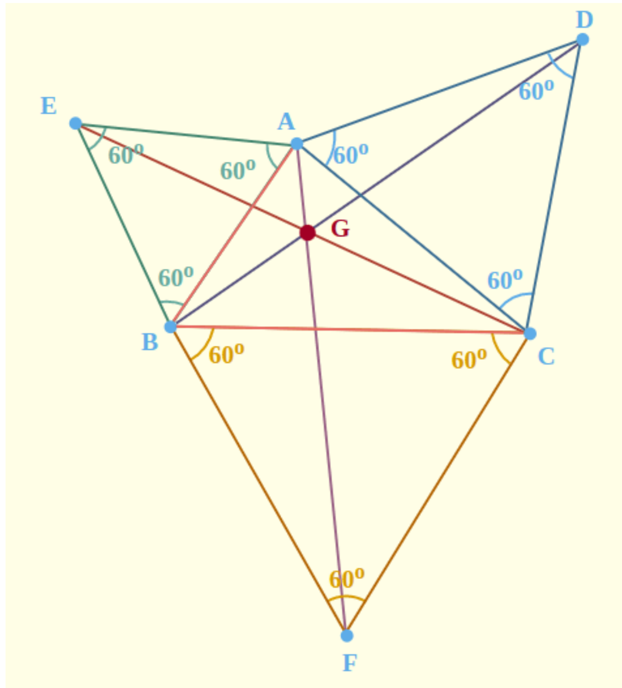


Figure 21: Visualising the rotation proof for the Fermat point. Rotating $\triangle APC$ by 60° to $\triangle AP'C'$ creates a straight line path $B - P - P' - C'$ when the distance is minimised.

3.3.3 Application: Robust Localisation in Sensor Networks

The mathematical property proved above has a direct application in Robust Statistics and Sensor Localisation.

In GPS-denied environments, a receiver often triangulates its position based on signal strength from three beacons (A, B, C).

- **The Problem:** Standard triangulation uses Least Squares (the Centroid), which is highly sensitive to outliers. If one beacon's signal is distorted by a wall (multipath interference), the Centroid shifts significantly, causing a localisation error.

- **The Solution:** The Fermat Point minimizes the L_1 norm $(\sum \|P - V_i\|)$ rather than the squared error (L_2). This makes it an outlier-robust estimator. Even if one beacon reading is extremely noisy, the Fermat point remains stable, providing a reliable location fix where the centroid would fail.

3.4 The Petr–Douglas–Neumann (PDN) Theorem

The Petr-Douglas-Neumann (PDN) theorem acts as the "Grand Generalization" of Napoleon's Theorem. It states that if one iteratively constructs specific isosceles triangles on the sides of *any* arbitrary n-gon, the sequence of shapes will eventually converge to a regular n-gon.

3.4.1 Relation to Napoleon's Theorem

This theorem is the direct iterative generalization of Napoleon's Theorem.

- **Napoleon's Theorem** is effectively the PDN process applied to a triangle ($n = 3$) for a single step ($k = 1$).
- **PDN Theorem** extends this logic to any polygon (n sides) and allows for multiple steps (k iterations).

Just as Napoleon filters out the irregularity of a triangle in a single step, PDN filters out the irregularity of an n-gon over multiple steps.

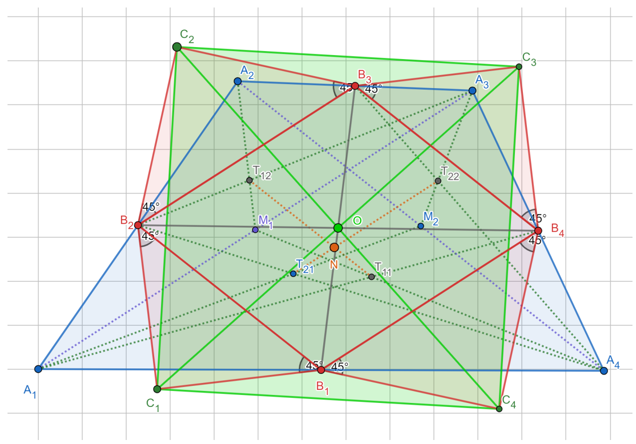


Figure 22: An illustration of the PDN iterative process, showing how an arbitrary initial polygon converges toward a regular polygon through successive transformations.
[**<empty citation>**]

3.4.2 Application: Shape Smoothing in Computer Vision

This computational model is mathematically identical to a Low-Pass Filter in signal processing, but applied to geometry.

- **LiDAR Data Cleaning:** Autonomous vehicles see the world as noisy polygons. A car detected by LiDAR often looks like a jagged, irregular shape due to sensor noise.
- **The PDN Filter:** By applying the PDN iteration to the raw detection data, the high-frequency jitter (noise) is dampened. The iteration effectively filters out the "irregular" eigenvalues of the shape matrix, leaving behind the smooth fundamental shape of the object.

3.5 Connections to the Course: Geometry for Computer Vision

A core principle of Geometry for Computer Vision is that geometric structure is characterised by invariants under appropriate transformation groups rather than by absolute coordinates. This viewpoint underpins all constructions and proofs in this project. Each proof begins by normalising the configuration using rigid motions, thereby removing translational and rotational degrees of freedom while preserving Euclidean invariants such as distances, angles, and incidence.

The vector and complex-number proofs of Napoleon's Theorem illustrate this approach in complementary ways. In the vector formulation, translating the triangle so that its centroid lies at the origin exploits translation invariance to simplify algebraic verification. In the complex formulation, rotations are encoded by multiplication with roots of unity, allowing planar isometries to be represented as scalar operations. Both techniques reflect standard course methods for representing geometric transformations as linear or algebraic operators.

Rotational isometries play a central role not only in Napoleon's Theorem but also in the construction of the Fermat–Torricelli point. In both cases, a $60°$ rotation converts a geometric or optimisation problem into one involving symmetry or collinearity. This demonstrates a recurring strategy emphasised in the course: apply an isometry to expose hidden structure while preserving the quantities of interest.

The contrast between Napoleon's Theorem and Thébault's Theorem highlights the distinction between Euclidean and affine invariants. Napoleon's construction holds for all triangles, whereas the quadrilateral analogue requires the input to be a parallelogram—precisely the class of quadrilaterals preserved under affine transformations. This reflects the weakening of invariants when passing from Euclidean to affine geometry, a theme repeatedly encountered in projective and affine models used in vision.

Finally, the Petr–Douglas–Neumann theorem exemplifies an iterative, transformation-based view of geometry. Repeated application of a fixed geometric operator drives an arbitrary polygon toward a regular configuration, analogous to geometric smoothing or normalization procedures in computer vision. Here, convergence reflects the dominance of invariant symmetric components under repeated transformation.

Overall, the results in this project reinforce a central methodological lesson of the course: geometric problems are most effectively addressed by selecting an appropriate transformation framework, reducing the configuration to a canonical form, and verifying invariants within that simplified setting.

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